

Reading Guide to the DROS Dual-Track Research Trajectory: 6-Paper Execution Program & Foundational Semantic Prehistory

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Abstract—This Technical Note provides an overarching reading guide to the DROS Dual-Track Governance Architecture, systematically unifying cognitive intent governance with downstream deterministic execution enforcement. It spans the semantic layer prehistory (Constraint-as-Code and DROS v7.3 DOR) for hallucination containment and manifestation isolation, alongside the 6-Paper Execution Program (6P, 4-Layer, PGM, WebMCP, Mobile, UAV) establishing binary in-band C-ABI gates, mobile OS privilege bounds, and drone kinetic invariants. Anchored by the open-source DROS-VEP Lite research vessel, this guide details the POSIX analogy, dual-throat abstractions, 6P baseline criteria, and authoritative Zenodo DOI citation identifiers across the entire research continuum.

Index Terms—Agent Runtime Governance, Dual-Track Architecture, Post-Compromise Security, Execution Authority, Physical AI, Evaluation Protocol.

I. Purpose and The 6 Research Scales

The DROS research trajectory expands a singular, foundational scientific question across six progressive scales of execution governance, supported by an open, reproducible evaluation vessel (DROS-VEP Lite):

$$\begin{array}{c} \text{Governance} \rightarrow \text{Runtime} \rightarrow \text{Kernel} \\ \rightarrow \text{Network (WebMCP)} \rightarrow \text{Digital} \rightarrow \text{Physical} \end{array} \quad (1)$$

These six scales construct an end-to-end causal defense chain:

- 1) Governance Model (6P): Enterprise trust delegation and revocation boundaries.
- 2) Runtime Architecture (4-Layer): Deterministic policy execution and forensic Merkle auditability.
- 3) Kernel Control (PGM): Post-compromise binary C-ABI microsecond hard tripping.
- 4) Network & Capability (WebMCP): Capability exposure, W3C delegation chains, and temporal closure in the Agentic Web.
- 5) Digital Substrate (Mobile): Privilege isolation across iOS/Android OS and protected APIs.
- 6) Physical Substrate (UAV): Kinetic envelope containment and lookahead braking in robotics.

II. The Overarching Research Thesis

$$\begin{array}{c} \text{Compromise(Cognitive Controller)} \\ \not\Rightarrow \text{Compromise(Downstream Execution Authority)} \end{array} \quad (2)$$

“A compromise of the autonomous cognitive controller does not inherently entail a compromise of downstream execution authority.”

Rather than relying exclusively on internal statistical alignment of generative models, the DROS research trajectory investigates the formal decoupling of cognitive intent from actual execution, establishing verifiable deterministic execution gates within declared enforcement boundaries that cannot be arbitrarily subverted by untrusted cognitive planes.

III. Dual-Track Decoupling: Semantic vs. Execution Governance

The DROS architecture formalizes a fundamental separation between cognitive intent alignment and downstream physical enforcement:

- Semantic & Epistemic Track (Constraint-as-Code & DOR): Governs internal cognitive content within domain-restricted knowledge systems through Vajra Contracts, flat-file decoupling, Deterministic Ontological Routing (DOR), and manifestation isolation to eliminate hallucination.
- Execution & Enforcement Track (The 6-Paper Program): Governs downstream effectful actions at binary C-ABI gates, capability bitmaps, and kinetic actuators to ensure unauthorized side-effects remain zero under compromise.

IV. Operating System Positioning & The POSIX Analogy

“DROS serves as a deterministic runtime governance substrate for autonomous AI workloads, establishing enforceable execution boundaries between cognitive agent intent and effectful execution.”

TABLE I
The DROS Research Trajectory: Core Technical Papers and Evaluation Vessel Matrix

Component	Paper / System Title	Role in Research Trajectory	Core Question / Functional Scope	Reference / DOI
Vessel	DROS-VEP Lite	Evaluation Protocol & Sandbox	How to provide an objective, falsifiable evaluation testbed?	GitHub Official Repo
Paper 1	DROS-6P	Governance Model	Who has authority? Why? When does it expire? (Enterprise Trust)	10.5281/zenodo.21833970
Paper 2	DROS 4-Layer	Runtime Architecture	How does policy deterministically reach execution? (Forensic Merkle)	10.5281/zenodo.22092008
Paper 3	DROS-PGM	Kernel-Level Control	Once compromised, can the execution control plane still block action?	10.5281/zenodo.21903687
Paper 4	DROS-WebMCP	Network Capability	How to govern WebMCP tool exposure with temporal closure in Agentic Web?	10.5281/zenodo.22290238
Paper 5	Post-Compromise Mobile	Digital Substrate	Does execution authority hold across Mobile OS/APIs?	10.5281/zenodo.22253147
Paper 6	Post-Compromise UAV	Physical Substrate	When actions cause physical motion, can the kinetic envelope hold?	10.5281/zenodo.22254372
Semantic 1	Constraint-as-Code	Foundational Paradigm	How to translate doctrinal classification into hard contracts with binary thread panic?	10.5281/zenodo.20823227
Semantic 2	DROS v7.3 (DOR)	Semantic Governance	How to deterministically constrain semantic drift and hallucinations in high-precision domains?	10.5281/zenodo.20776075
Foundational 1	Runtime Attribution	Attribution Archetype	How to resolve MAS context-blindness with non-repudiable Ed25519 forensic evidence?	10.5281/zenodo.20823163
Foundational 2	Vajra Sandboxing	Dual-Isolation Archetype	1,000 adversarial tests: can binary C-FFI tripping neutralize bypasses of semantic layers?	10.5281/zenodo.20823189

Analogous to POSIX: POSIX did not eliminate application complexity; it prevented every application from reinventing low-level OS interfaces. DROS applies the exact same abstraction principle to agent governance—upstream applications and cognitive agents need not independently invent identity, authorization, tool boundaries, gating, forensic auditability, and microsecond revocation mechanisms. Special Note: This analogy describes an architectural role; it does not claim DROS replaces general-purpose operating systems like Linux or Windows.

V. Minimal Dual-Throat Interface: Ingress & Egress

To upstream applications, the ideal architecture merely routes through a data Ingress and an action Egress; cryptographic DIT tokens, capability bitmap checks, Merkle audit chains, and RCU revocations are encapsulated by the DROS substrate:

The relationship between DROS and application-layer governance middleware (e.g., policy SDKs) is summarized in the 4-Layer paper:

They decide. DROS enforces.

Application layers declare high-level semantic intent and policies; unmanaged native execution paths must still be deterministically constrained at binary C-ABI / FFI boundaries.

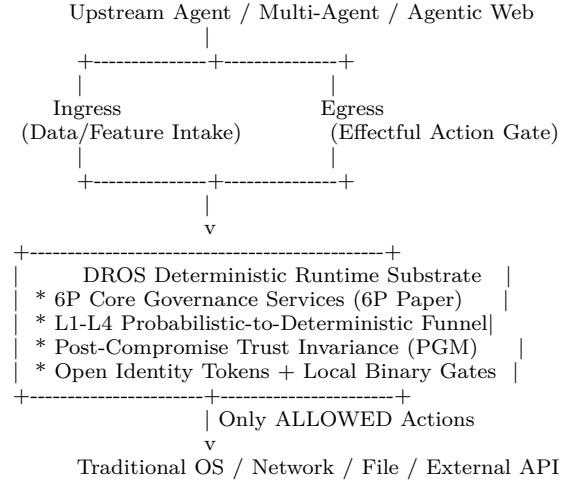


Fig. 1. Minimal Dual-Throat Abstraction and Runtime Substrate

VI. 6P: The Minimum Operating Baseline for Agent Governance

The series asserts that for autonomous agents to act as accountable execution entities in regulated environments, the six dimensions in Table II represent the minimum operational baseline for governance completeness:

VII. Empirical Metrics & Epistemic Interpretation

Reported latencies, interception rates, and ablation ratios must be interpreted strictly within their declared threat models and measurement segments:

TABLE II
DROS-6P Core Runtime Governance Service Matrix

Dimension (Pillar)	Core Mechanism & Direction (See 6P Paper)
Principal	Cryptographic Delegated Identity Tokens (DIT) resolving context-blindness
Authorization	Deterministic capability bitmaps for $O(1)$ in-memory enforcement
Tool Bound	Binary in-band interception constraining tool execution boundaries
Policy Gate	Dynamic PII masking, HITL triggers, and selective disclosure
Audit Log	Immutable SHA-256 Merkle hash chains for forensic legal evidence
Revocation	Atomic pointer swaps (RCU) enabling instantaneous sub-microsecond revocation

- 1) Measurement Segmentation: Policy evaluation latency ($\approx 26 \mu s$) and C-ABI binary gate latency (< 500 ns) measure distinct code segments and must not be conflated into a single end-to-end figure.
- 2) Conditional Invariance: Statements asserting “100% interception” apply strictly within the declared threat models, evaluated datasets, and harness configurations, rather than as unconditional guarantees against arbitrary unregistered binary vulnerabilities or total OS kernel compromise.
- 3) Ablation Implication: Ablation experiments in the 4-Layer paper demonstrate that absent L4 binary gating, adversarial obfuscations can penetrate into unauthorized execution; this reinforces the thesis that semantic defenses and execution enforcement are complementary and mutually indispensable.

VIII. Scope of Guarantee & Research Boundaries (Mandatory)

The DROS series provides deterministic execution governance and execution-boundary enforcement; it explicitly does not guarantee:

- Internal cognitive alignment or statistical non-hallucination of generative models;
- Security over arbitrary unregistered native binaries or fully compromised operating system kernels;
- Automated moral judgment over authorized-in-scope but contextually improper business actions;
- Complete regulatory certification (technical evidence supports forensic compliance, but organizational and procedural controls remain essential).

IX. Substrate vs. Protocol Positioning (The VEP Vessel)

This series formalizes the research problem as an executable, attackable, and measurable experimental substrate problem:

- 1) VEP Research Vessel (Evaluation Protocol): Formalizes vendor-neutral RFC-001 Execution Governance Evaluation Standards, providing multi-architecture baselines, 72-hour soak tests, and an open falsification channel.

- 2) DROS Runtime (Reference Substrate): Operates as a concrete, deterministic binary reference implementation under the VEP specification.

- 3) Dual-Domain Instantiations (Digital & Physical):

- Mobile: Evaluates Unauthorized Protected System-Effect Invariance ($\text{Auth} = 0 \implies \text{PSE} \equiv 0$) and microsecond revocation races ($T_{\text{rev}} < 2.5 \mu s$).
- UAV: Evaluates Unauthorized Command Invariance ($\text{Auth} = 0 \implies \Delta u_{\text{cmd}} \equiv 0$) and physical safety-envelope preservation ($\mathcal{S}_{\text{safe}}$).

X. Cross-Domain Substrate Mapping

DROS demonstrates that deterministic runtime governance applies uniformly across heterogeneous operational substrates:

- Digital Domains (Mobile OS): Protects privileged APIs, clipboard, storage, and sensors against unauthorized state leakage.
- Physical Domains (Robotics & UAVs): Enforces physical kinetic invariance, dynamic deceleration horizons, and geofence boundary containment.

XI. Recommended Reading Paths

- Architects & Decision Makers: Start with this Guide $\rightarrow$ explore the DROS-VEP Lite Root README.
- Foundational Paradigm & Cognitive Alignment: Constraint-as-Code $\rightarrow$ DROS v7.3 (DOR Framework) $\rightarrow$ DROS 4-Layer (L1/L2 Semantic Ingress).
- Foundational Archetypes & Inception: Runtime Attribution $\rightarrow$ Vajra Sandboxing $\rightarrow$ DROS-6P $\rightarrow$ DROS 4-Layer.
- Governance & Policy Researchers: DROS-6P $\rightarrow$ DROS 4-Layer.
- Security Engineers & Binary Specialists: VajraClaw Archetype $\rightarrow$ DROS-PGM $\rightarrow$ Post-Compromise Mobile.
- Robotics & Control Theorists: Post-Compromise UAV (Kinematics, braking horizon, latency decoupling).

XII. References & Identifiers

When referencing technical claims, cite via the official DOI or technical report identifier:

- 1) DROS-VEP Lite: Verifiable Execution Protocol & Evaluation Sandbox, 2026. <https://github.com/Top-Celestial-Company-Ltd/DROS-VEP-lite>
- 2) DROS-6P: DOI: [10.5281/zenodo.21833970](https://doi.org/10.5281/zenodo.21833970)
- 3) DROS 4-Layer: DOI: [10.5281/zenodo.22092008](https://doi.org/10.5281/zenodo.22092008)
- 4) DROS-PGM: DOI: [10.5281/zenodo.21903687](https://doi.org/10.5281/zenodo.21903687)
- 5) DROS-WebMCP: DOI: [10.5281/zenodo.22290238](https://doi.org/10.5281/zenodo.22290238)
- 6) DROS-Mobile: DOI: [10.5281/zenodo.22253147](https://doi.org/10.5281/zenodo.22253147)
- 7) DROS-Kinetic (UAV): DOI: [10.5281/zenodo.22254372](https://doi.org/10.5281/zenodo.22254372)
- 8) Constraint-as-Code: DOI: [10.5281/zenodo.20823227](https://doi.org/10.5281/zenodo.20823227)
- 9) DROS v7.3 (DOR): DOI: [10.5281/zenodo.20776075](https://doi.org/10.5281/zenodo.20776075). Repo: <https://github.com/Top-Celestial-Company-Ltd/Dharma-Reasoning-Operating-System>
- 10) Runtime Attribution: DOI: [10.5281/zenodo.20823163](https://doi.org/10.5281/zenodo.20823163)
- 11) Vajra Sandboxing: DOI: [10.5281/zenodo.20823189](https://doi.org/10.5281/zenodo.20823189)